

CHINMAY MURTHY

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ECE student specializing in **autonomy, sensor fusion, and real-time systems**. Proven leader in hardware/software integration, delivering high-performance algorithms in C++/Python for industry and research. Passionate about owning complex technical challenges from embedded edge devices to full-stack fleet management.

EDUCATION

University of Washington

Seattle, WA

Bachelor of Science, Electrical and Computer Engineering (3.7) | Music Minor

Sep 2022 – Jun 2026

- Relevant Coursework: Machine Learning, Computer Architecture, Embedded Systems, Signal Processing, Data Structures & Algorithms, Control Systems

EXPERIENCE

Software Engineering Intern

June 2025 – Sep 2025

Bidgely

Palo Alto, CA

- Performed **supervised fine-tuning** of a large language model (LLM) on a curated **domain-specific dataset**
- Designed and implemented a **custom retrieval-augmented generation** (RAG) pipeline for proprietary data, reducing **response latency by 47%**

Undergraduate Research Assistant

Oct 2024 – May 2025

University of Washington

Seattle, WA

- Optimized **sensor fusion** algorithms (MSCKF-based VINS) leveraging **computer vision** and GPS data for **autonomous** outdoor ground robots

Geegah LLC

Ithaca, NY

Engineering Intern

June 2024 – Sep 2024

- Brought up and verified **RF circuitry** in next-gen (4x pixel density) ultrasonic imager with 200+ components
- Reengineered **FPGA** architecture for **25% lower resource use** and **100x performance gain**
- Created Python API for USB communication, enabling successful **customer demos** and **IEEE IUS** presentation

Applications Intern

June 2023 – Sep 2023

- Integrated proprietary imager with 3axis motion control stage presented at **DARPA ERI summit 2023**
- Prototyped fluidics, electronics, and full-stack web app for water quality PoC with a **Python/Flask REST API**

EXTRACURRICULARS AND PROJECTS

Controls Software Lead

Sep 2022 – Present

Advanced Robotics at the University of Washington (ARUW)

Seattle, WA

- Designed and implemented **Cascade PID-LQG controller** of a **6DOF balancing two-wheel-legged robot** (5-bar active suspension), **simulated in Matlab/Simulink** before **STM32** deployment
- Implemented **inverse kinematics** and gravity compensation for **mobile 6DOF differential manipulator**
- Engineered an **autonomous path planner** that can navigate and follow **dynamic obstacles** using **LiDAR**
- Developing a **Meta-Imitation Learning** pipeline distilling **RL** and **MPC** controllers into a **recurrent foundation policy** for **online system identification** and **zero-shot adaptation**
- Managed **team of 10** to deploy a **75k+ LoC C/C++** codebase, delivering a **2.5 m/s autonomous path-planning** algorithm and a **Kalman Filter/CV sensor fusion** pipeline achieving **2-3 cm drift** over 100m

Control Tower (Fleet Management System) | React, Python, GitLab, sCons

May 2025

- Engineered a real-time **fleet management dashboard** processing **500Hz telemetry** and aggregating live network metadata from distributed **Raspberry Pi** and **Nvidia Jetson** compute nodes
- Architected a **dynamic IP resolution** database and **API**, integrating with **sCons** to enable **automated firmware flashing** to microcontrollers based on real-time hardware target data

Smart Twister | ESP32, React

May 2025

- Developed a novel optimization approach to combat pressure sensor cross-talk, and a full-stack web interface.

SKILLS

Languages: C/C++, Python, Java, Verilog/SystemVerilog, Matlab, TypeScript

Tools: Git, React, OpenCV, Matlab/Simulink, Unix/Linux, ROS